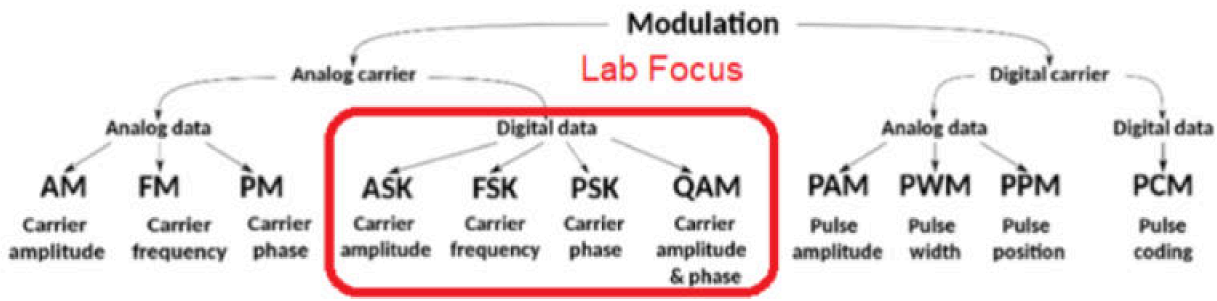
Digital Phase Modulation Lab



Introduction and Lab Summary

This lab focused on the concepts explained below in this summary. We reviewed basic digital modulation signals (ASK, FSK and PSK) on a scope and spectrum analyzer and explored advanced phase modulation signals (QPSK, 8PSK) and QAM (4, 8, and 16QAM) signals on a scope and were introduced to IQ mapping and PN data patterns. A sine wave has three properties we can change to carry information: amplitude, frequency, and phase. Amplitude is the height of the wave and represents how strong or loud the signal is. Frequency is how fast the wave oscillates (how many cycles per second). Phase is the position or direction the wave starts from. In digital modulation we change only one of these properties to represent digital data (0s and 1s).

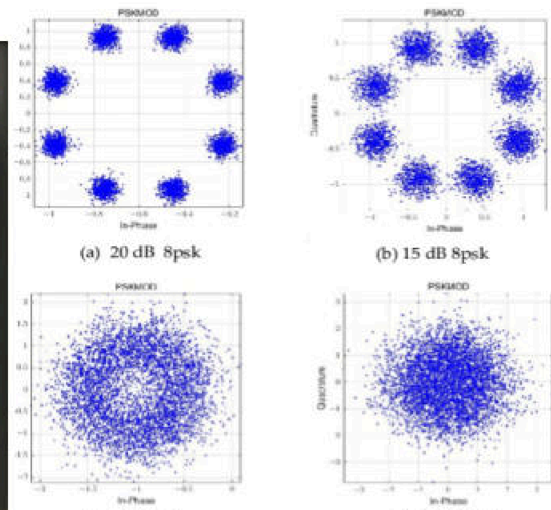
Amplitude Shift Keying (ASK) is the digital equivalent of AM, where the height of the carrier wave is modulated/ changed by the digital data. A digital "1" is transmitted using a strong, tall signal, while a digital "0" uses a weak or even no signal. Frequency Shift Keying (FSK) is similar to FM and sends bits by changing the frequency of the signal (the modulation frequency fm). A "1" uses a fast, high-frequency wave, while a "0" uses a slower, low-frequency wave. Phase Shift Keying (PSK) sends data by changing the starting direction of the wave. In the simplest version (Binary PSK or BPSK), a "1" uses a normal sine wave while a "0" uses a 180° phase-shifted sine wave that is flipped upside down.PSK is the most important of the three methods because it is more resistant to noise. Noise can easily distort amplitude or frequency which can cause errors in ASK and FSK. However, phase changes tend to remain clearer even in noisy environments. This is why PSK is widely used in modern digital communication systems (like WIFI or cellular networks).

IQ modulation is a way of sending digital data by changing the direction, or phase, of a radio wave. Instead of only turning a signal on and off or changing how fast it wiggles IQ modulation splits the data into two parts. The two parts are I (in-phase) and Q (quadrature). The I part is lined up with a normal sine wave, and the Q part is lined up with a cosine wave (which is just the same wave shifted by 90 degrees). These two pieces are then added together to make one signal that can point in different directions depending on what bits we want to send.. If both I and Q represent the digital value “1” then both waves are added together and the result points in a new direction (a new phase) such as 45° ($\text{Sin}(\omega t) + \text{Cos}(\omega t) = \text{Sin}(\omega t+45$ degrees) or alternatively $\text{Cos}(\omega t-45)$. If the data bit is “0”, that means we would get the inverse signal out of the mixer and the output would be waves flip upside-down before being added and the result points in a different direction like 225°: $-\text{Sin}(\omega t) - \text{Cos}(\omega t) = \text{Sin}(\omega t+225$ degrees).

IMG 1. *In a simple 4-point system like 4-QAM or QPSK, there are four dots placed around a circle, each about 90° apart (for example: 45°, 135°, 225°, and 315°). Each dot in the constellation represents one possible combination of the I and Q signals and therefore one possible phase the wave can take. This pattern (from top right and CCW Counter- Clockwise) is 1) Sin(ωt+45degrees), 2) Sin(ωt+135 degrees). 3) Sin(ωt+225degrees), or Sin(ωt+180degrees+45degrees) and finally 4) Sin(ωt+45degrees +270degrees), which is Sin(ωt+315degrees) which is also Sin(ωt-45degrees).*

Each dot stands for a different bit pattern, so when the receiver sees which dot the signal lands closest to, it knows which bits were sent. We can draw all the possible directions the signal can point on a graph called an IQ map, where each dot shows one allowed signal state. Each dot stands for a specific pattern of bits- for example BPSK has two dots (two phases) that send one bit at a time. QPSK has four dots and sends two bits at a time. 16-QAM has sixteen dots and sends four bits at a time. The more dots there are, the faster we can send data (more bits per symbol and hence a higher data rate), but it also becomes more sensitive to noise.

IMG 2.



(a) 20 dB 8psk (b) 15 dB 8psk (c) 5 dB 8psk (d) 0 dB 8psk

IMG 3.

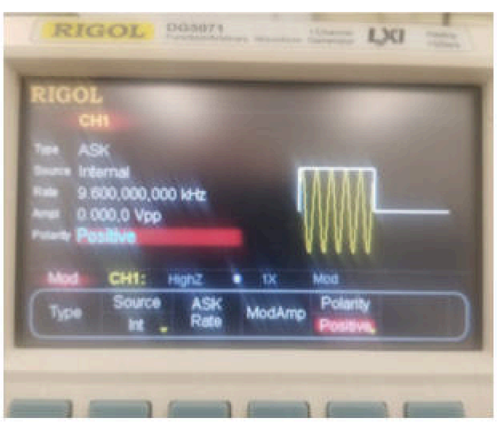
To read the IQ map in image 1, for example for 16 PSK (16 phase shift keying) there are 16 dots and each dot represents one allowed phase the carrier wave can take when transmitting data. We can see that the data rate (how fast the digital data is being sent) is 9.6kpbs or 9,600 bits per second. Because there are 16 different phase positions, each point can carry 4 bits of information (because 2^4 = 16). Therefore 0 degrees would be 0000, 45 degrees would be 0010, etc. Therefore instead of sending one bit at a time like BPSK, 16 PSK sends 4 bits at a time.The horizontal line is the I axis (in phase) and the vertical line is the Q axis (quadrature), and every dots angle from the center is the phase of that signal state. When a signal is transmitted, the received point should land on or close to one of these dots. If a point lands near the right most dot, it indicates a phase near 0 degrees. The top most dot is a phase near 90 degrees, left most 180 degrees, and so on.

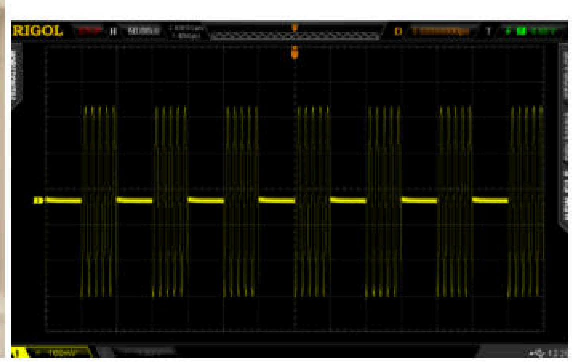
It is theoretically possible to have 32 PSK and even higher, but noise becomes an issue.Noise as a random signal behaves like a vector with any phase, magnitude or length, determined by the signal/ noise ratio of the system. In image 2, we can see what happens to a 8 PSK signal when noise increases. In 8-PSK there are eight dots arranged around a circle (as seen in image 2) and each dot represents a different phase direction that carries 3 bits of information (because 2^3=8). In the first image (where the signal to noise ratio is 20 dB) the noise vectors start to blur or fuzz the star pattern- but as long as there is a clear area separating each star, the receiver should still be able to determine each of the 8 individual signals. This means almost no bit errors.In the second image (15 dB signal to noise ratio), the added noise is enough that separating the individual 8 signals without error would be difficult. In the third image (5 dB signal to noise ratio), the dots are very fuzzy and are starting to overlap, and it is clear that the 8 distinct stars are not discernable and therefore errors would likely occur in the receiver.This shows that as noise increases, the constellation points spread out and PSK becomes unreliable because the dots are too close together and too easy to confuse. To help with noise, we can move some points inward toward the center instead of having everything on the same circle. This creates a new type of constellation with different amplitudes as well as phases which is the idea behind QAM (Quadrature Amplitude Modulation). QAM helps maintain separation between dots even in noise, allowing higher-bit modulation (more data per symbol) than PSK alone can handle without excessive errors.

ASK

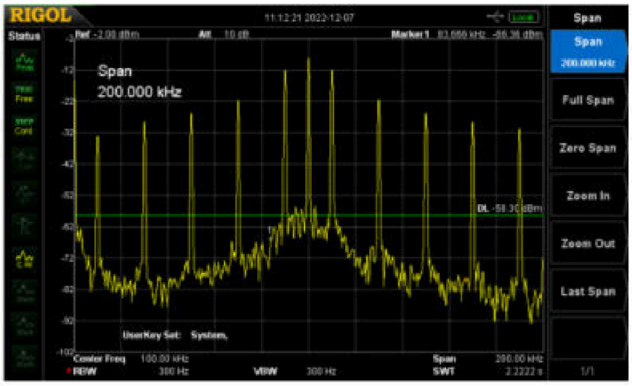
1. First, to review ASK we set the carrier frequency by selecting Sine and set the frequency to 100kHz, and 1Vpp. 100kHz is not special but we used it as 10x the modulation frequency.

2. Press MOD button then TYPE (far left button)
3. Select ASK for Amplitude Shift Keying
a. Set the ASK Rate to 10kHz; 1/10 of the carrier.
b. Set the ModAmp level to 0Vpp
We used the Sync signal from our Rigol generator as an external input to your scope to trigger this signal. We noticed that there are 5 cycles on and 5 cycles off since the carrier is 10X the Modulation.On the spectrum Analyzer (Span 200kHz) we saw the carrier and many sidebands as each harmonic in the 10kHz modulation frequency showed up. We needed to set our RBW (Resolution BW) on the Spectrum Analyzer to 1kHz or 300Hz to see this.

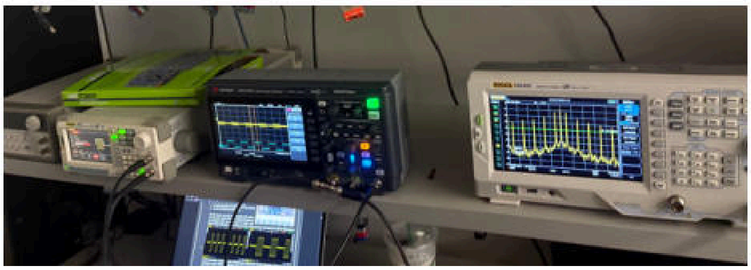




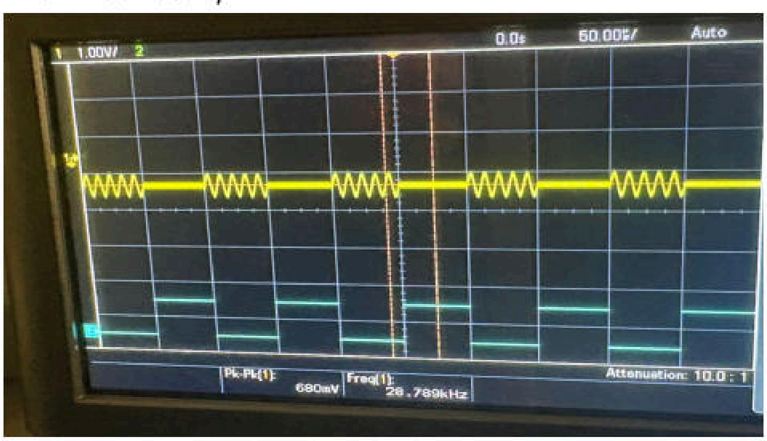
IMG 4. *Set up* **IMG 5.** *Expected results*



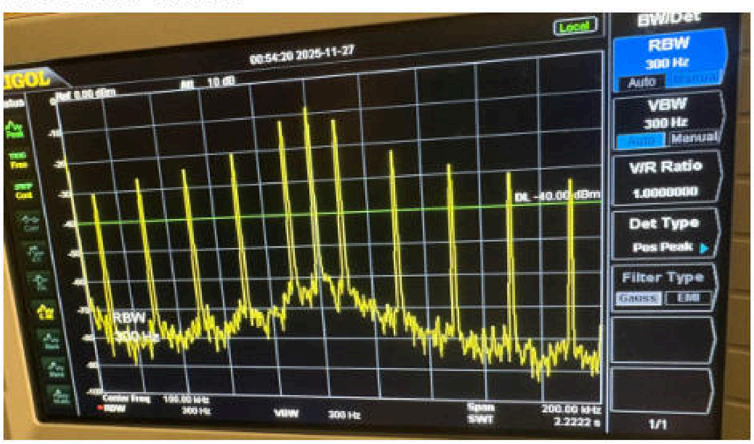
IMG 6. *Expected results*



IMG 7. *Our set up*



IMG 8. *Our results*



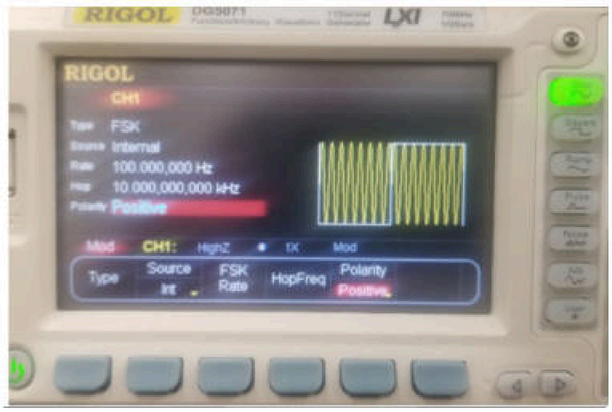
IMG 9. *Our results*

FSK

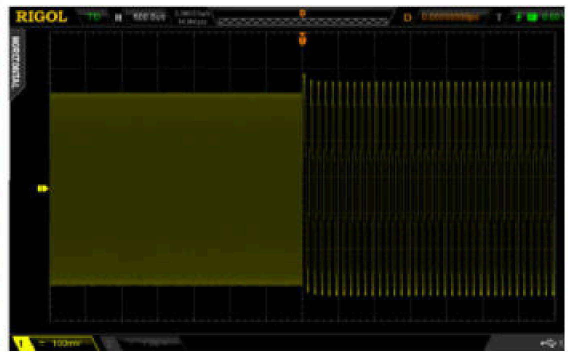
To switch to FSK

1) Press MOD button 2) under Type Select FSK for Frequency Shift Keying where the FSK Rate to 100Hz; hop or shift is 10kHz 3) use the Sync signal from your Rigol generator as an external input to your scope to trigger off of this signal. Notice the high and low FSK frequencies and the Modulation and some amplitude distortion.

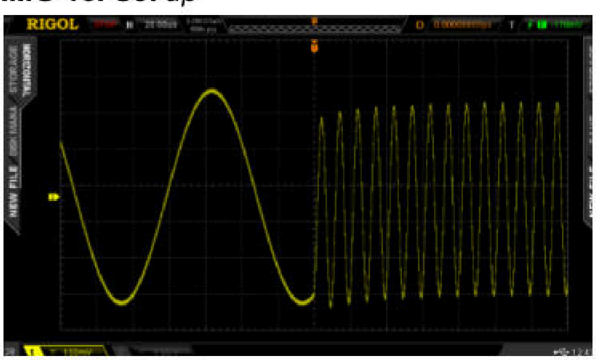
On the spectrum Analyzer we saw the carrier and many sidebands as there are many Bessel J terms involved related to the 10kHz hop frequency. For this display to show up we had to set our RBW (Resolution BW) on your Spectrum Analyzer to 1kHz or less.



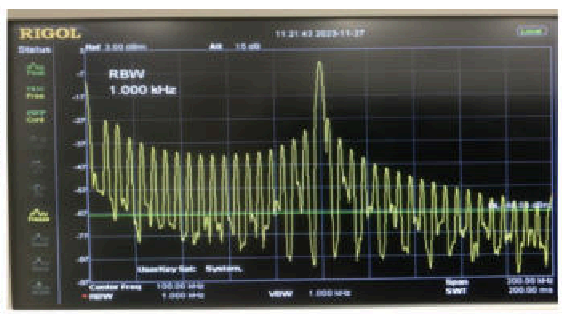
IMG 10. *Set up*



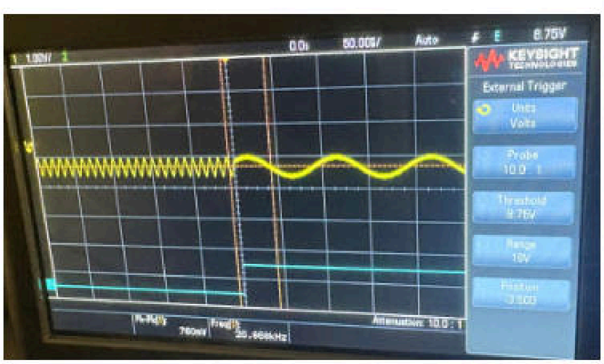
IMG 11. *Expected results*



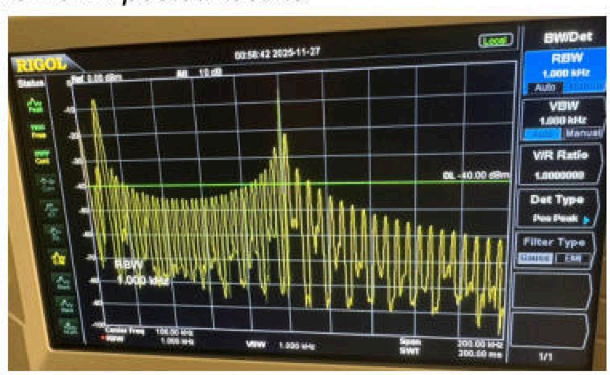
IMG 12. *Expected results*



IMG 13. *Expected results*



IMG 14. *Our results*

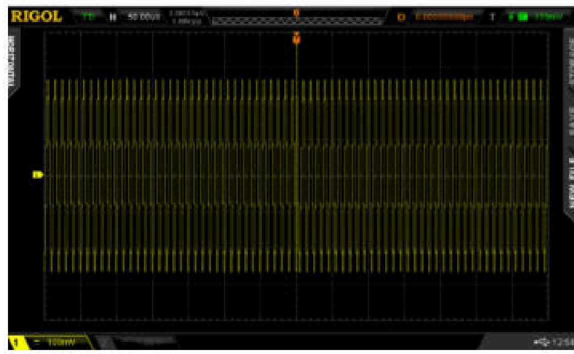


IMG 15. *Our results*

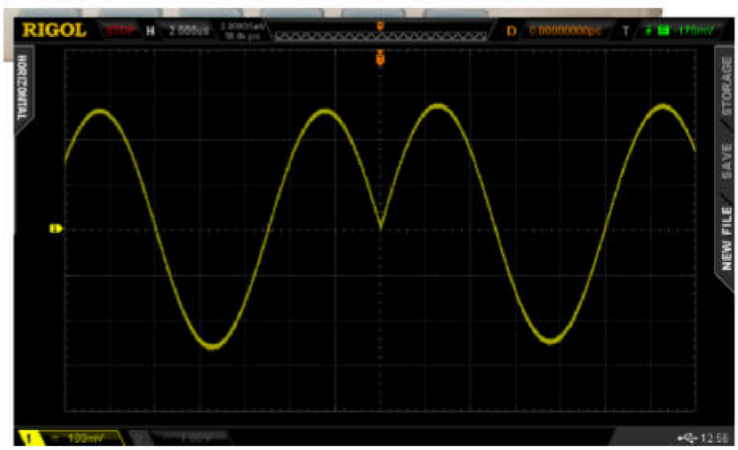
PSK

To switch to PSK we first pressed MOD button then Type and Select PSK for Phase Shift Keying. We set the PSK Rate to 100Hz; Phase (change) is 180. This is also called BPSK or binary PSK.

IMG 16. *Set up*

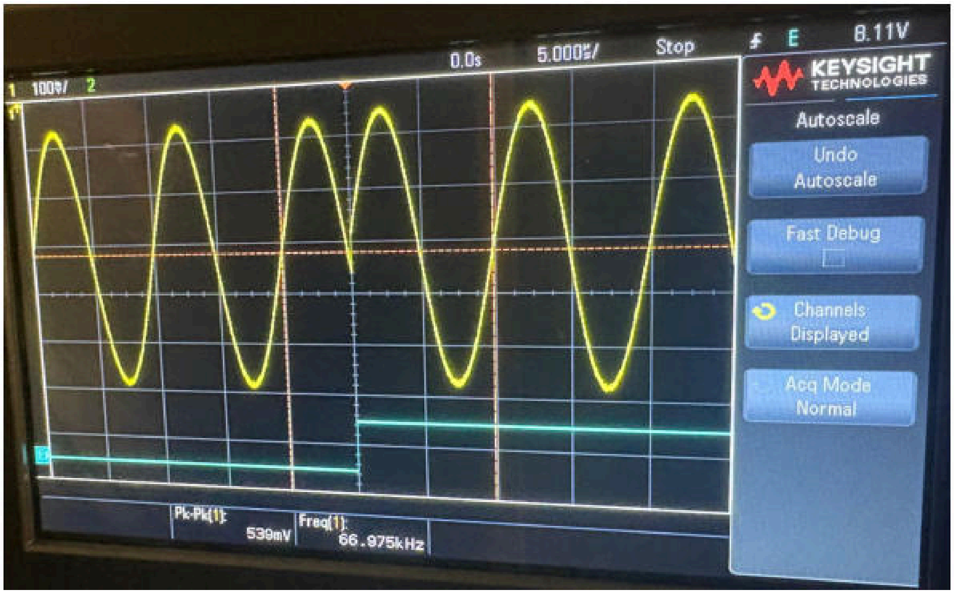


IMG 17. *Expected results*

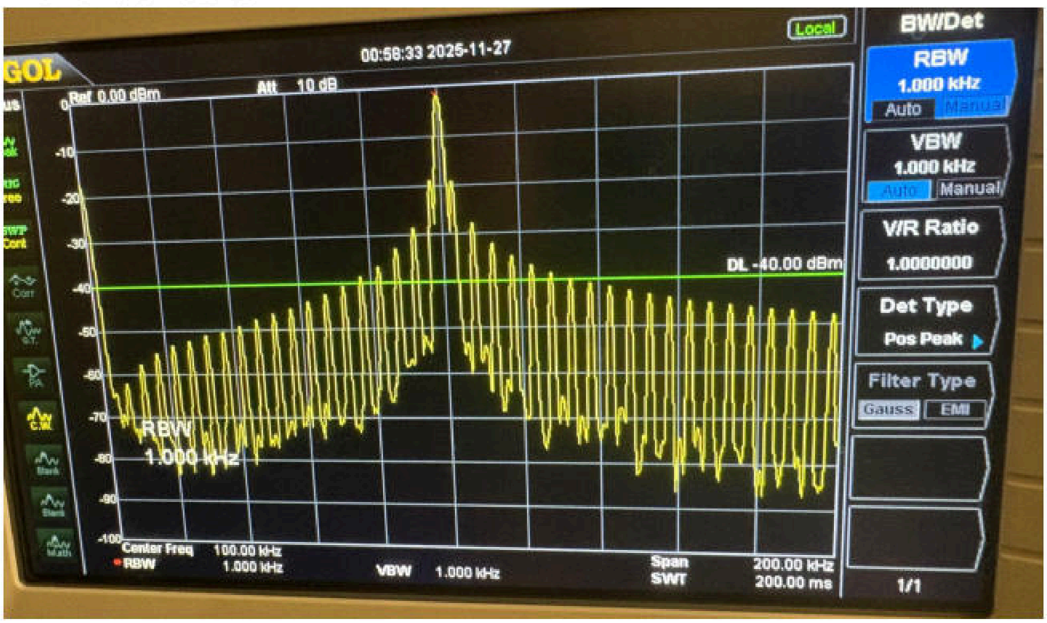


IMG 18. *Expected results*

IMG 19. *Expected results*



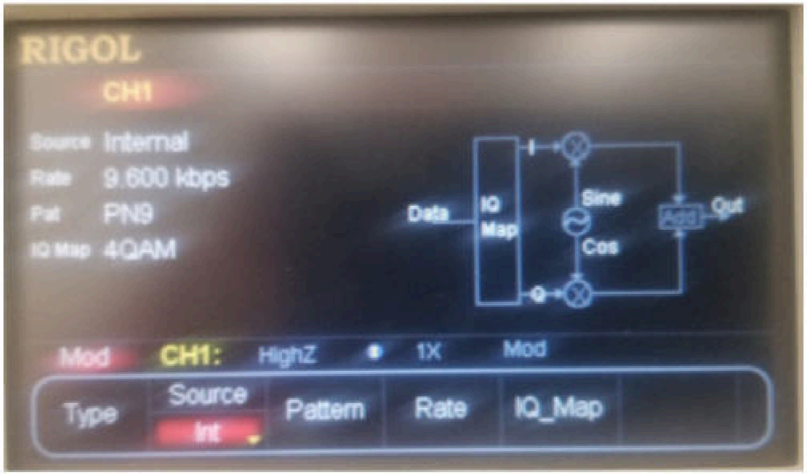
IMG 20. *Our results*



IMG 21. *Our results*

Advamced cases of PSK: IQ and QAM/ QPSK

Then, we pressed the MOD button again and TYPE. We noticed the red arrow > just to the right of the PSK selection. This > arrow indicates there is an extended menu accessed by using the right > and left < arrow buttons just below and to the right of the PSK indicator. After selecting the right arrow > we saw the PWM and IQ selections. Only IQ is selectable, PWM is greyed out because it only works with a digital square wave carrier, which was done last week, so we selected IQ.

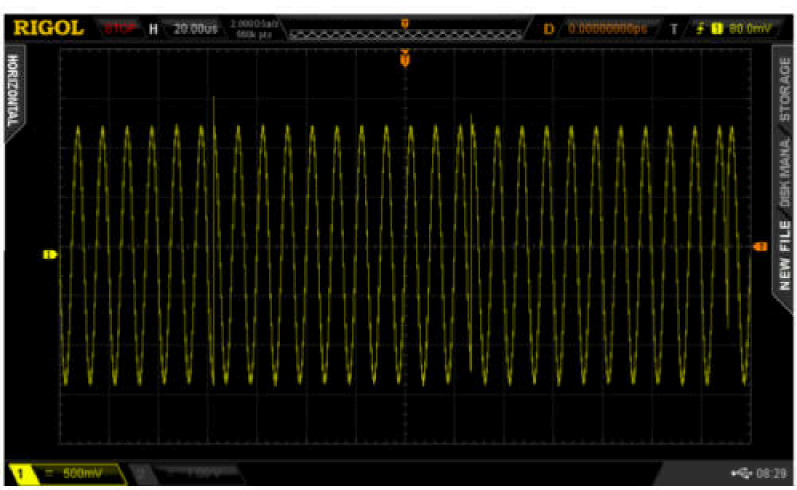


IMG 22. *IQ modulation diagram*

After selecting IQ one sees a block diagram which is referred to as an IQ modulation diagram. In the diagram Digital Data enters the IQ_Map block from the left. The IQ map block creates both an I and Q digital output or data stream. For more information about the concept of and reading IQ maps, the lab summary at the beginning of this lab report should be referenced.

To start the lab measurements, select Mapping Select and select BPSK. BPSK is an abbreviation for Binary Phase Shift Keying. As the IQ_Map and constellation diagram indicates there are only two possible signal states with an In-Phase component- at 0 degrees or its inverse at 180 degrees. Although the Q component is not used in this case the IQ diagram representation is still typical for BPSK digital modulation.

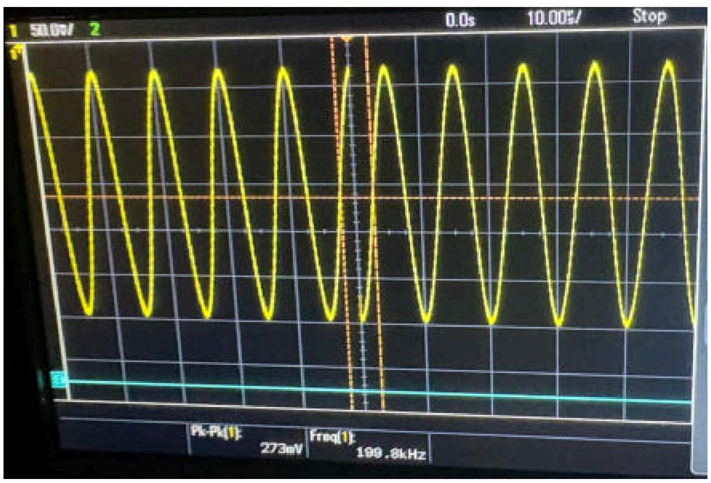
When viewed on the scope (use the SINGLE button to capture one set of data sweep) we saw the below BPSK signal.

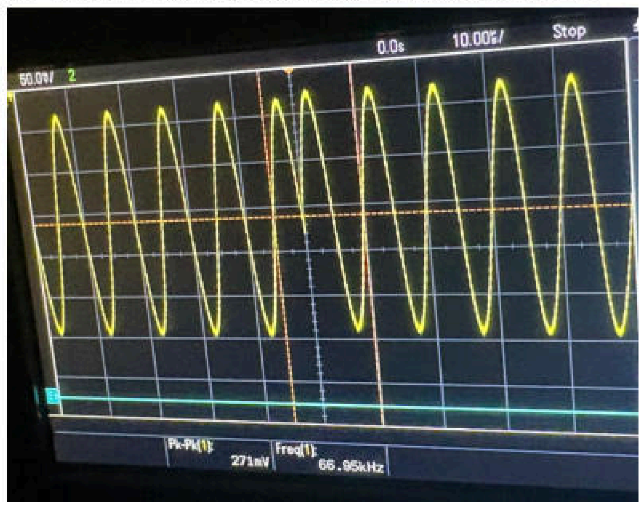


IMG 23. *Expected*

and

IMG 24. *Notice that the above sinewave signal is modulated or changes about every 5 horizontal divisions which is related to the Rate setting. A spike or vertical line are indications of those data transition points. If we zoom into the 1st and 2nd these are the changes we see*



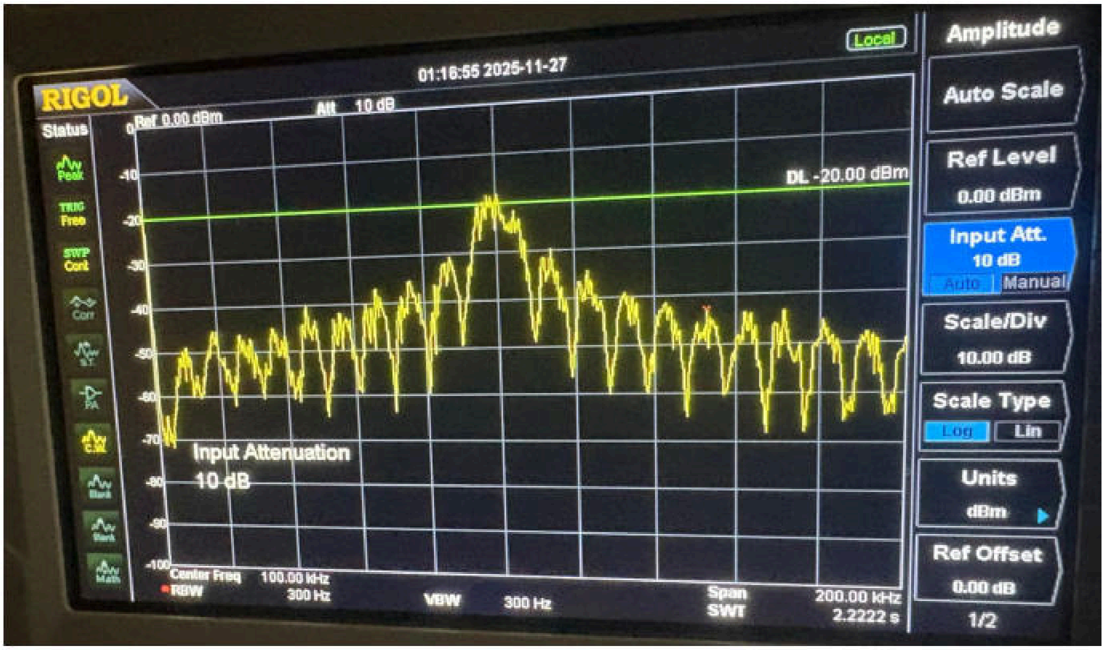


IMG 24. *Our results*

IMG 25. *Our results*

Examining the phase change for each of the above cases we can see that a phase change of 180 degrees occurred at each point, since BPSK only has two cases- 0 degrees and 180 degrees. PSK with 180 degree modulation is the same as BPSK. The switching spikes seen above are related to the higher data rate of 9.6kbps (9600 bits per second) compared to the previous PSK modulation frequency of 100Hz.

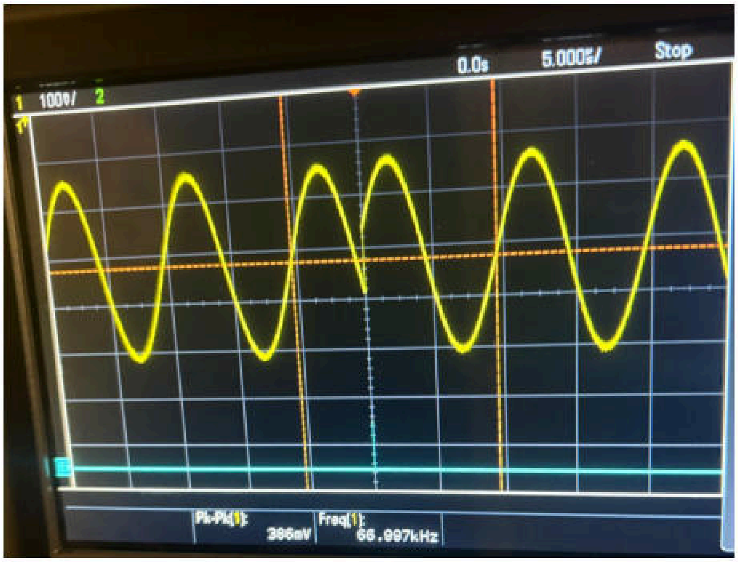
The BPSK frequency spectrum (shown in image 26) is not easy to interpret. In general, digital modulation signals tend to be more useful in the time domain.



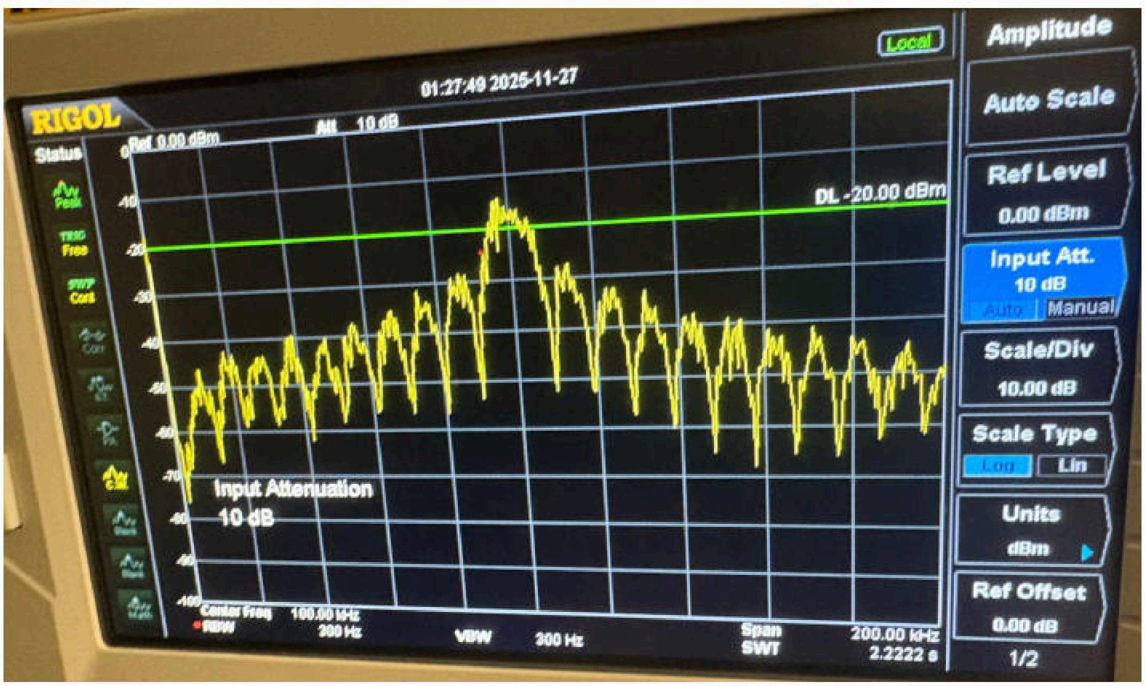
IMG 26. *Our results*

Going beyond simple BPSK, the next level of PSK is 4PSK. This is generally referred to as QPSK or quadrature PSK, as it has four states. This can be seen in the IQ map in IMG 27. The constellation diagram shown is identical to the previous 4QAM diagram. Indeed QPSK and 4QAM are equivalent in the signals created but will utilize different demodulation receiver setups

IMG 27. *4PSK*



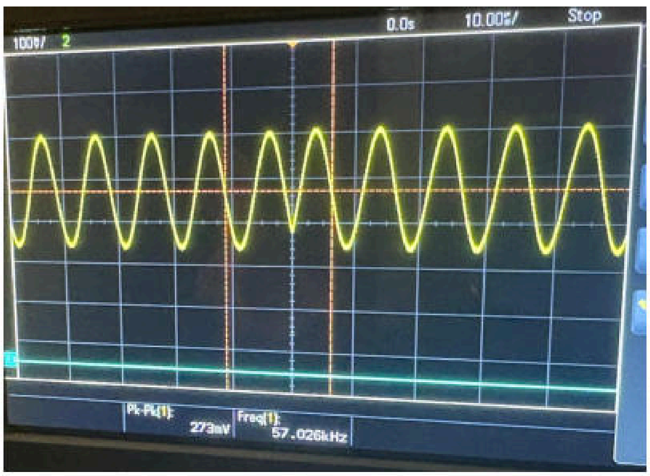
IMG 28. *Our results*



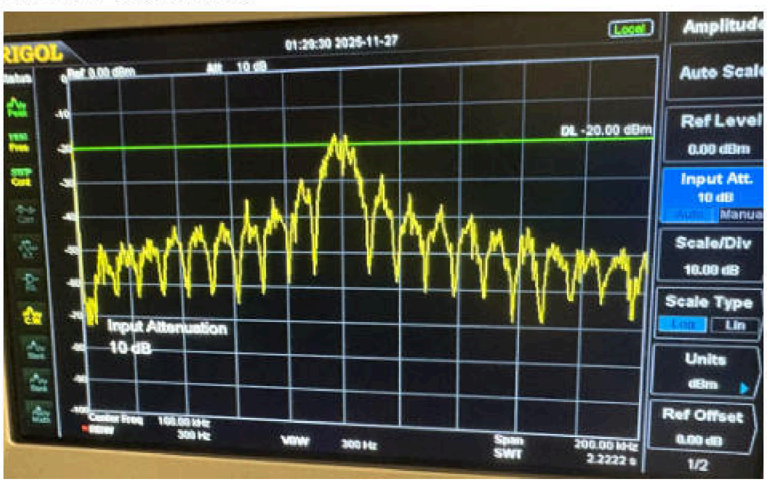
IMG 29. *Our results*

Beyond QPSK is 8PSK using the Mapping Select. As expected, the IQ constellation diagram now shows 8 separate phase points in a circle pattern. Each signal represents a phase shift of 45 degrees (i.e. 360 degrees /8), as seen in image 30.

IMG 30. *8PSK*



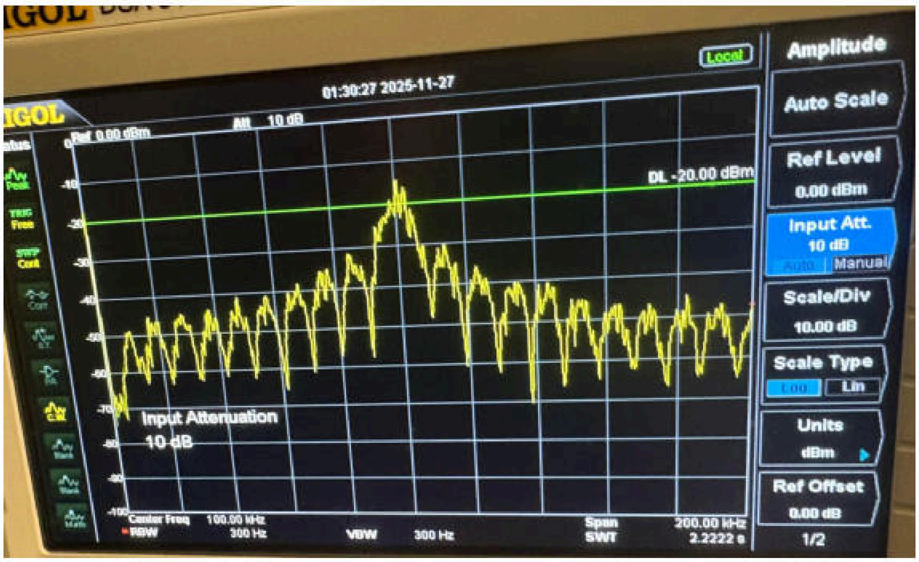
IMG 31. *Our results*



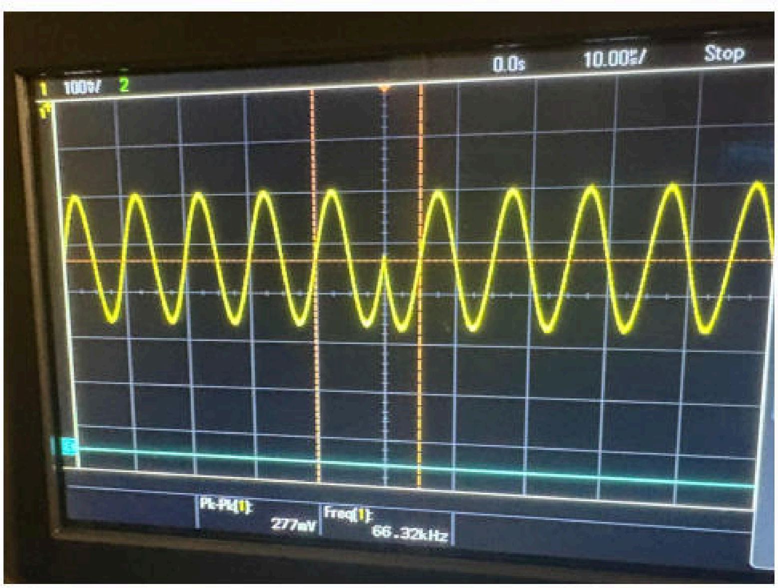
IMG 32. *Our results*

For 16PSK, as expected the IQ Constellation diagram shows components around a unit circle separated by 22.5 degrees (i.e. 360 degrees/16)

IMG 33. *8PSK*



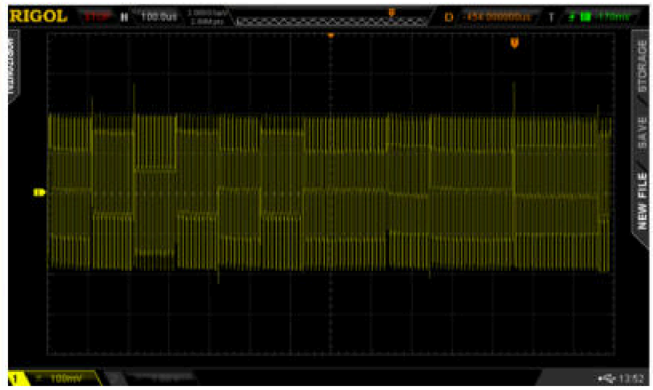
IMG 34. *Our results*



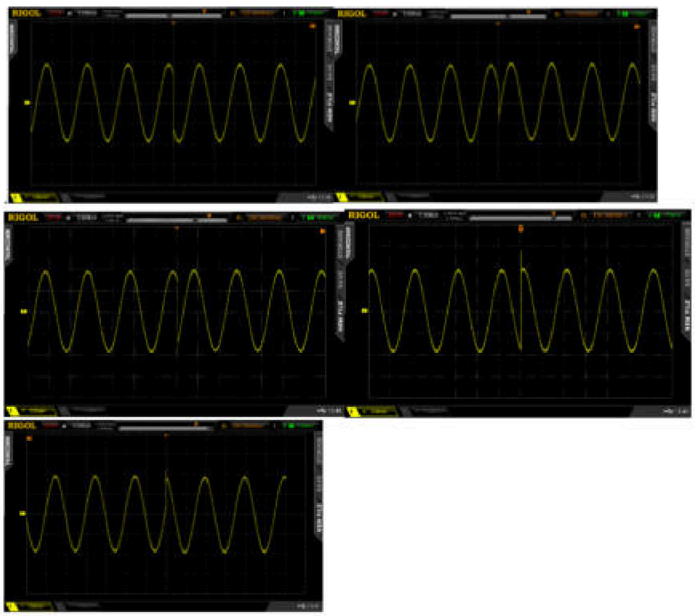
IMG 35. *Our results*

Returning back to the MODulation option, we selected 4QAM and reviewed its constellation diagram. This constellation diagram should look familiar as it is identical to the QPSK diagram asbthe two signals QAM and QPSK are the same thus the time domain views should be the same as shown.

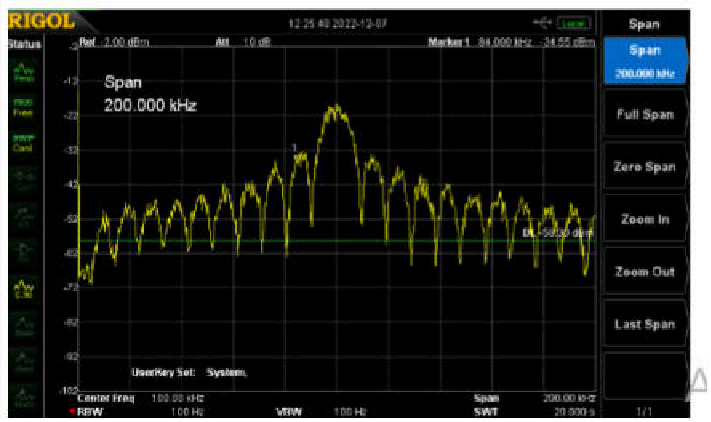
IMG 36. *4QAM*



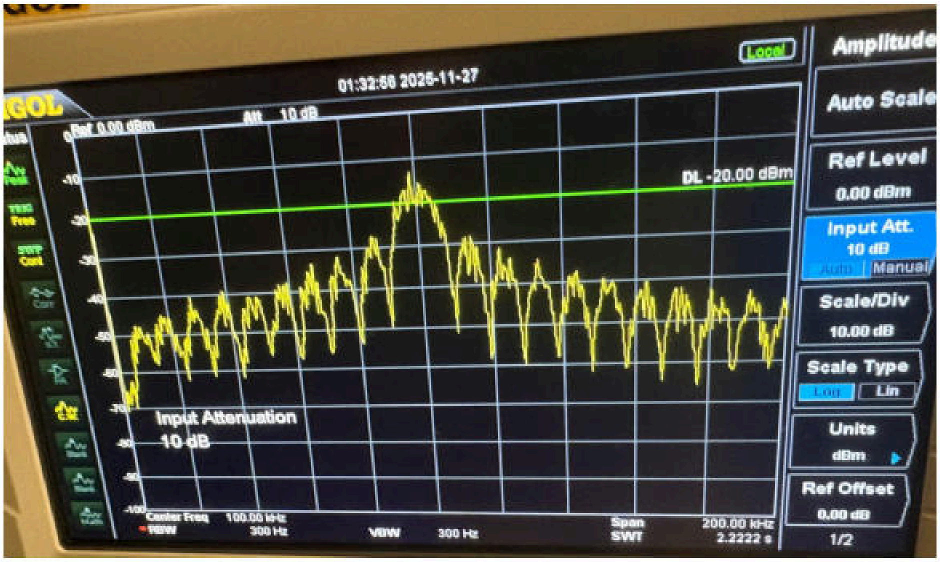
IMG 37. *Expected*



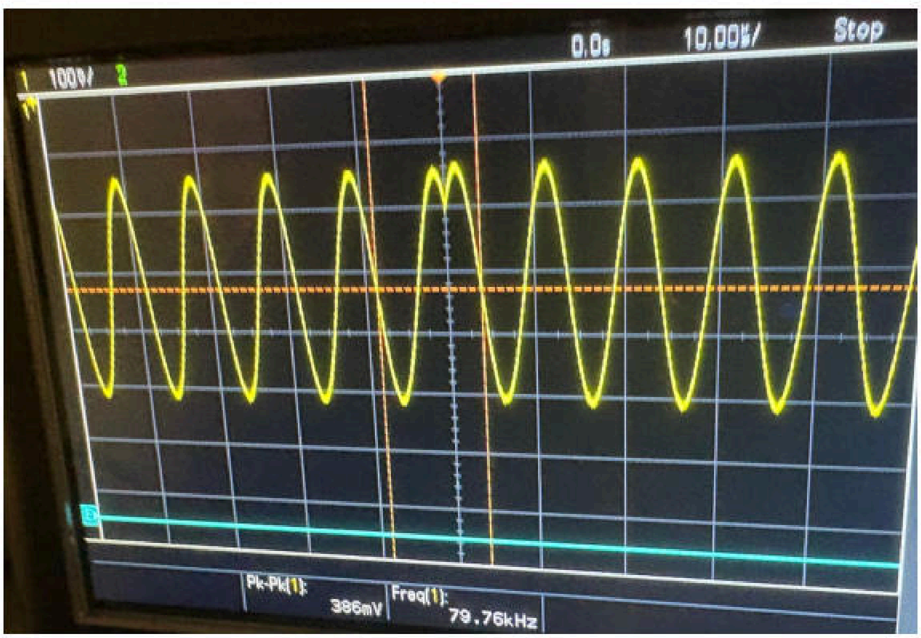
IMG 38. *Expected*



IMG 39. *Expected*



IMG 40. *Our results. The frequency spectrum is more complex, but one can definitely see the SINC (Sinx/x) pattern characteristic of a digital or pulse type pattern.*

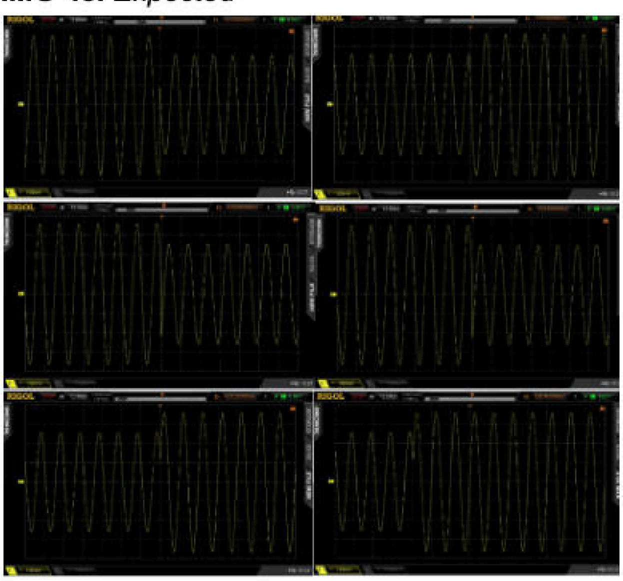


IMG 41. *Our results*

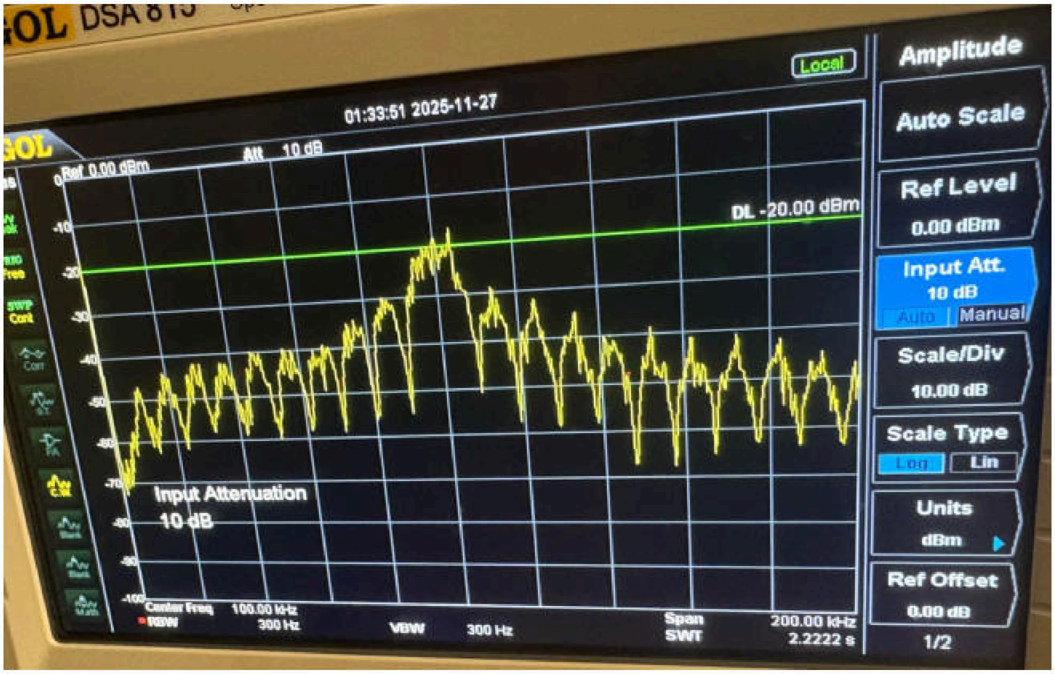
When we go back to MOD and select 8QAM a new type of constellation diagram is shown. Besides locations at every 45 degrees one also sees terms at 0 degree, 90 degrees, 180 degrees and 270 degrees. These new vectors also have a different length from the center of plot indicating that the amplitude of such signals will be different than the terms from the 4QAM/QPSK constellation diagram. As expected two positive amplitudes and 2 related negative amplitudes are appear corresponding to these 8QAM signa

IMG 42. *8QAM*

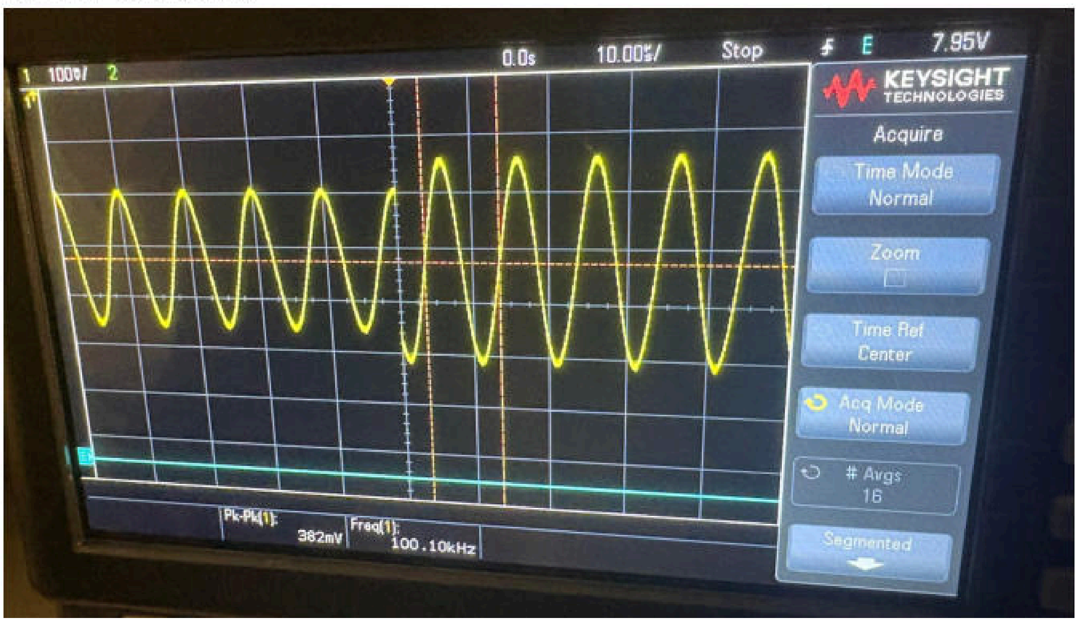
IMG 43. *Expected*



IMG 44. *Expected*



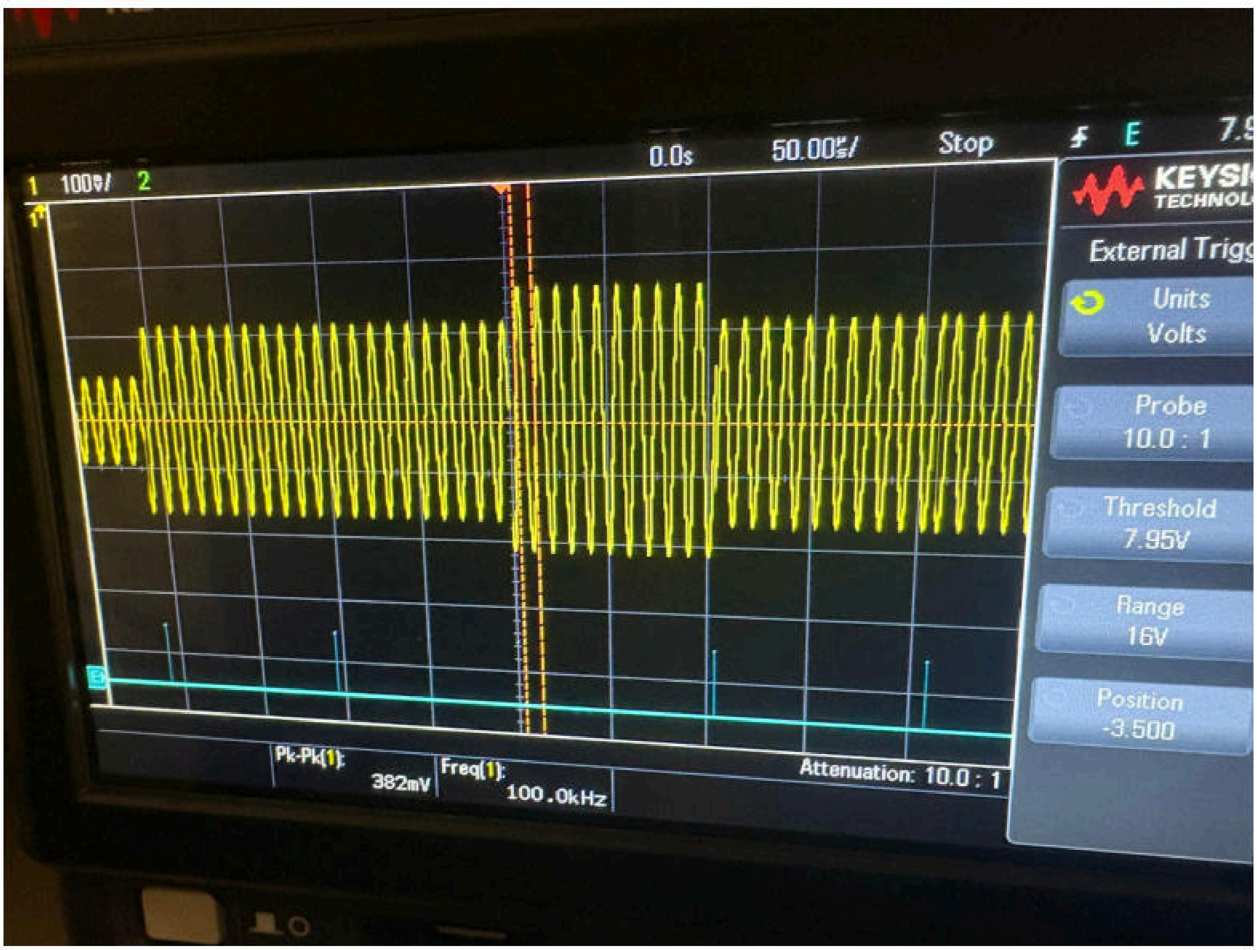
IMG 45. *Our results*



IMG 46. *Our results*

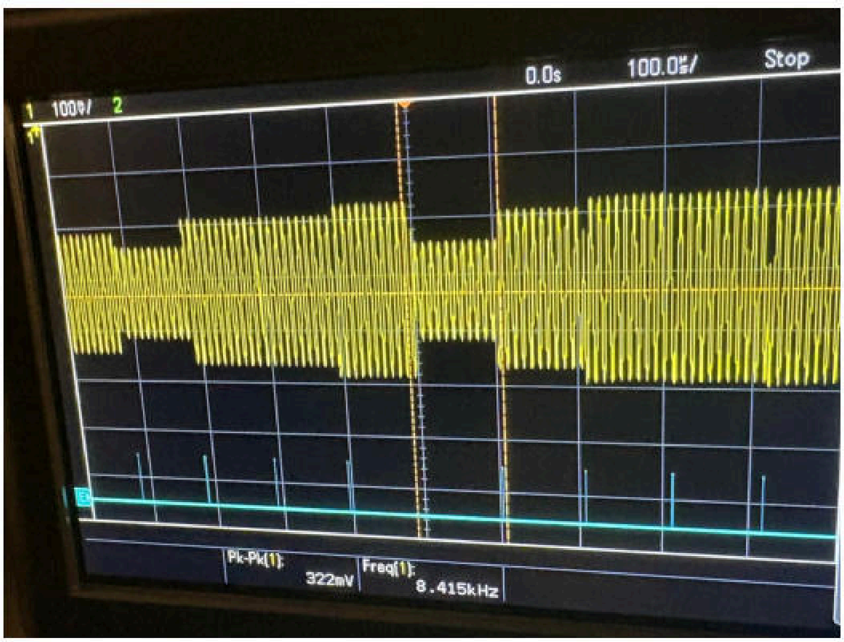
Beyond 8-QAM, there are higher-order schemes such as 16-QAM, 32-QAM, and 64-QAM, each with their own IQ constellation diagrams containing more dots (more possible signal states). By looking at one of these higher-order QAM signals on the oscilloscope we can understand what is happening in the time domain. The location of each dot in the constellation shows the exact phase and amplitude being transmitted at that moment.. When the symbol changes from one dot to another the phase and/or amplitude of the actual waveform changes and these transitions appear as spikes or sudden jumps in the time-domain signal. The cleaner the transitions and the closer each received point is to its ideal dot, the easier it is for the receiver to correctly identify which symbol (and therefore which bits) were sent.

IMG 47. *16QAM*



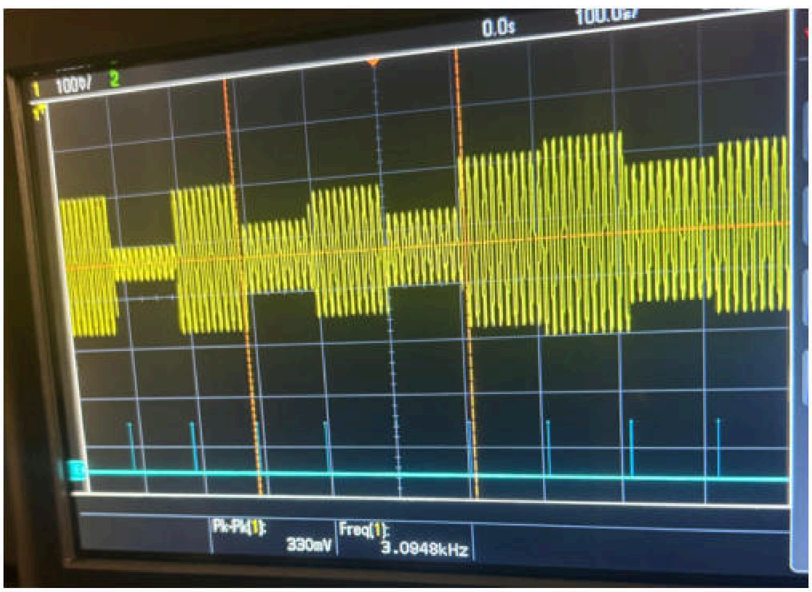
IMG 48. *Our results*

IMG 49. *32 QAM*



IMG 50. *Our results*

IMG 51. *64QAM*



IMG 52. *Our results*